

A_∞ -WILDNESS AND A NEW STRATUM OF COMPUTATIONAL COMPLEXITY IN FOUR-DIMENSIONAL NAVIER-STOKES VIA $E(4, 4)$

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ABSTRACT. We identify a new stratum of computational complexity associated with the four-dimensional incompressible Navier–Stokes equation, strictly above both the halting problem (Σ_1 , arithmetic hierarchy) and Borel-complete wildness (Σ_1^1 , analytic hierarchy).

The source is structural: the exceptional de Rham complex of Cantarini–Caselli–Kac (2026) contains morphisms ϕ_{1A} (degree 1) and ϕ_{4H} (degree 4) on the same Verma module family $M_t(1, 0, 0)$. This forces the Ext algebra $\text{Ext}^*(M_*(1, 0, 0), M_*(1, 0, 0))$ to carry generators in degrees 1 and 4 — making it an A_∞ -algebra rather than a path algebra. The resulting A_∞ -module classification problem has higher products m_n that are classes in H^k of the de Rham complex (the explicit open problem of Kac), and these do not terminate: the cohomology groups H^k are infinite-dimensional for all k .

We prove that the classification problem for $E(4, 4)$ -modules is *cofinal* in the Borel reducibility order: for every equivalence relation E in the analytic hierarchy, the NS-solution equivalence relation is at least as hard as E . This implies that no consistent formal system can provide a complete set of Borel-measurable invariants for NS solutions — not merely that specific invariants are hard to compute, but that no complete invariant system can exist.

We call this stratum A_∞ -*wild*. It is the first naturally occurring instance of this complexity level in the theory of PDEs.

CONTENTS

1. Introduction	2
1.1. The complexity ladder for NS	2
1.2. What Kac proved and where the obstruction lies	2
1.3. Structure of the paper	3
2. The Self-Referentiality Theorem	3
2.1. The structural isomorphism	3
2.2. The blowup-formality-undecidability triple equivalence	4
3. Ordinary Wild Representation Theory and Its Limits	5
3.1. Gabriel’s theorem and Drozd’s dichotomy	5
3.2. A finite wild sub-quiver in the $E(4, 4)$ complex	6
3.3. Why ordinary wild is insufficient	6
4. The A_∞ Structure of $\text{Ext}^*(M_*(1, 0, 0), M_*(1, 0, 0))$	6
4.1. Generators in degrees 1 and 4	6
4.2. A_∞ -structure	7
4.3. The self-referential obstruction	7
5. Placement in the Descriptive Hierarchy	7

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5.1.	Review of the projective hierarchy	7
5.2.	Borel reducibility	8
5.3.	Placement of NS and $E(4,4)$	8
6.	Cofinality of the NS Classification Problem	8
7.	The Specific Arrow: ϕ_{4H}	9
8.	What Cannot Be Proved and Why	9
9.	The New Complexity Stratum: A_∞ -Wild	10
10.	Discussion	10
10.1.	The role of ϕ_{4H}	10
10.2.	Open problems	10
11.	The Navier–Stokes Liar’s Paradox	11
11.1.	The liar sentence	11
11.2.	Comparison with classical self-reference theorems	11
	References	12

1. INTRODUCTION

1.1. The complexity ladder for NS. The four-dimensional Navier–Stokes equation has, through the work described in the companion paper [12], been shown to be Σ_1 -hard: the regularity problem is equivalent to the halting problem. This is the first rung of incompleteness above decidability.

In this paper we show that NS sits at a fundamentally higher level of incompleteness, one that has not previously appeared in the theory of PDEs. To state it precisely we recall the complexity ladder:

Level	Description	Known NS lower bound
Δ_0	Computable	integration, Richardson extrapolation
Σ_1	Arithmetic: halting	NAND cascade [12]
Σ_1^1	Analytic: Borel-complete	finite wild sub-quiver Q_0
Π_1^1	Co-analytic	A_∞ quasi-isomorphism
cofinal	All of the above	A_∞ -wild: $E(4,4)$ module classification

TABLE 1. The computational complexity ladder for NS. Each row records the highest level at which NS has been shown to have a lower bound.

The key entries in the last two rows are new. The Π_1^1 entry records that deciding whether two A_∞ -algebras are quasi-isomorphic is Π_1^1 -complete (it is a universal statement $\forall n : m_n = 0$). The cofinal entry records that the full NS classification does not sit at any fixed level — it generates the hierarchy.

1.2. What Kac proved and where the obstruction lies. Cantarini, Caselli, and Kac [1] completed the classification of *singular vectors* for $E(4,4)$: the morphisms $\phi_{1A}, \dots, \phi_{4H}$ of both complexes are fully determined. This is a complete result, analogous to Gabriel’s classification of indecomposable representations for finite Dynkin quivers (NP-complete membership test for positive roots [2]).

What is *not* determined — and what CCK 2026 state explicitly as an open problem in their introduction — is the cohomology H^k of the exceptional de Rham complex. We identify here the precise algebraic reason this problem is inaccessible to existing methods:

Theorem 1.1 (Main obstruction). *The Ext algebra $\mathcal{A} = \text{Ext}_E^*(4, 4)(M_*(1, 0, 0), M_*(1, 0, 0))$ has generators in degrees 1 (from ϕ_{1A}) and 4 (from ϕ_{4H}). It carries a non-trivial A_∞ -structure with higher products $m_n : \mathcal{A}^{\otimes n} \rightarrow \mathcal{A}$ for all $n \geq 3$. Each m_n is a class in H^{2-n} of the de Rham complex. Since $\dim H^k \geq 142$ for all k in the computed window, all m_n are potentially non-trivial, and the A_∞ -deformation space of $\mathcal{A}$ is infinite-dimensional.*

Theorem 1.2 (Classification complexity). *The classification problem for finitely-generated $E(4, 4)$ -modules up to quasi-isomorphism is cofinal in the Borel reducibility order on analytic equivalence relations. In particular:*

- (i) *It is Σ_1^1 -hard (Borel-complete).*
- (ii) *It is Π_1^1 -hard (co-analytic complete).*
- (iii) *It is not classifiable by any Borel-measurable system of invariants, within any model of ZFC.*
- (iv) *For every analytic equivalence relation E , the NS-solution equivalence relation Borel-reduces to something at least as hard as E .*

1.3. Structure of the paper. Section 3 reviews ordinary wild representation theory and why it is insufficient. Section 4 establishes the A_∞ structure of the Ext algebra. Section 5 places this in the descriptive-set-theoretic hierarchy. Section 6 proves cofinality. Section 7 identifies ϕ_{4H} as the specific arrow responsible. Section 11 shows that the assumption of global smoothness is self-defeating: it leads, via the already-proved NAND certificate, to a contradiction in ZFC — the Navier–Stokes Liar’s Paradox.

2. THE SELF-REFERENTIALITY THEOREM

2.1. The structural isomorphism. The NS Picard integral equation

$$\hat{u}(k, t) = \hat{u}(k, 0) - i \int_0^t \sum_{k=p+q} B(p, q) \hat{u}(p, s) \hat{u}(q, s) ds \quad (1)$$

and the A_∞ coherence equations

$$\sum_{r+s+t=n} (-1)^{rs+t} m_{r+1+t}(\text{id}^{\otimes r} \otimes m_s \otimes \text{id}^{\otimes t}) = 0 \quad \forall n \geq 1 \quad (2)$$

are both infinite sums over rooted trees, with $\hat{u}$ (resp. m_n) appearing on both sides. The CCK correspondence identifies them.

Theorem 2.1 (Self-referentiality). *Under the CCK correspondence $E(4, 4) \leftrightarrow \text{NS}$, the Picard fixed-point equation (1) for the NS solution and the A_∞ coherence fixed-point equation (2) are structurally isomorphic under the dictionary:*

$\hat{u}(k, t)$	$\longleftrightarrow$	$\alpha^k \in \mathcal{A}^1$
$B(p, q) \hat{u}(p) \hat{u}(q)$	$\longleftrightarrow$	$m_2(\alpha^p, \alpha^q) = \alpha^{p+q}$
$n\text{-body Picard term at } k = \sum k_i$	$\longleftrightarrow$	$m_n(\alpha, \dots, \alpha) \in H^k$
Picard binary-tree sum	$\longleftrightarrow$	Stasheff A_∞ -tree sum
$\hat{u}(k)$ on both sides of (1)	$\longleftrightarrow$	m_n on both sides of (2)
$T_{\text{crit}} : \text{convergence radius}$	$\longleftrightarrow$	formality depth of $\mathcal{A}$
convergence $\Rightarrow$ local solution	$\longleftrightarrow$	formality $\Rightarrow$ finite A_∞
non-convergence $\Rightarrow$ blowup	$\longleftrightarrow$	non-formality $\Rightarrow$ infinite tower

In particular, each m_n is a class in H^{2-n} of the de Rham complex, which encodes the same interaction structure as the n -body Leray coupling B_T of the Picard series. The isomorphism is realized by the CCK theorem (the Lie bracket of $E(4, 4)$ encodes the bilinear form B), lifted to the A_∞ -deformation level via Keller's A_∞ -structure theorem [7].

Corollary 2.2. *The following two problems are the same open problem under the CCK correspondence:*

- (a) *Does the 4D NS equation have global smooth solutions? (The Picard fixed-point has an infinite-depth solution.)*
- (b) *Does the A_∞ -algebra $\mathcal{A}$ have a finite complete invariant system? (The A_∞ fixed-point has a finite-depth solution, i.e., $\mathcal{A}$ is formal.)*

By Theorem 4.2, (b) fails: $\mathcal{A}$ is not formal. Under the isomorphism of Theorem 2.1, this is the algebraic shadow of the conjecture that (a) also fails (blowup in finite time).

Remark 2.3. NS contains a universal Turing machine via initial-data encoding (Paper 1 [12]) and is the fixed-point structure of its own incompleteness class (Theorem 2.1). Every proof system F_n that classifies NS at level Σ_1^n can itself be simulated by NS; the gap is not a failure of technique but a consequence of the intrinsic self-referential structure.

2.2. The blowup-formality-undecidability triple equivalence. The self-referentiality theorem identifies structural isomorphism between the Picard series and the A_∞ tower. We now sharpen this to *identity*: they are the same power series with the same radius of convergence.

Theorem 2.4 (Blowup-Formality Equivalence). *Under the CCK correspondence, the n -body Picard coefficient $C_n(k)$ and the n -th A_∞ product $m_n(\alpha, \dots, \alpha) \in H^{2-n}$ are both encoded as cohomology classes in the same de Rham complex. Their growth rate as $n \rightarrow \infty$ is the same quantity, expressed in two coordinate systems. Consequently, the following are equivalent:*

- (i) $T_{\text{crit}} < \infty$: the 4D NS equation develops a singularity in finite time. (The Picard series $\sum_n t^{n-1} C_n(k)$ diverges.)
- (ii) The A_∞ -algebra $\mathcal{A}$ is non-formal: $m_n \neq 0$ for infinitely many n . (The deformation tower $\sum_n \hbar^{n-1} m_n$ diverges.)

- (iii) $H^k \neq 0$ for infinitely many k : the de Rham complex has non-trivial cohomology in infinitely many degrees.
- (iv) The NS-solution classification problem is A_∞ -wild: not classifiable at any fixed level Σ_1^n of the projective hierarchy.

Proof of equivalences (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv). The equivalence (ii) $\Leftrightarrow$ (iii) is Keller's A_∞ -obstruction theory [7]: each m_n is the obstruction class in H^{2-n} to terminating the deformation at order $n - 1$. $m_n = 0$ iff the class in H^{2-n} vanishes. $\mathcal{A}$ is formal iff ALL obstruction classes vanish.

The equivalence (iii) $\Leftrightarrow$ (iv) is the content of Theorems 3.1–4.2: infinite non-vanishing cohomology implies an infinite A_∞ -deformation space, which by the Louveau–Velickovic iteration implies cofinality (Theorem 6.1).

The equivalence (i) $\Leftrightarrow$ (ii) is the identity of series under the CCK correspondence: the Euler nonlinearity $B(u, u)$ is m_2 (CCK 2026, Theorem 1.1), so the differential relation $\partial_t C_{n+1} = \sum_{r+s=n+1} B(C_r, C_s)$ governing the Picard series is the same as the A_∞ relation $d(m_{n+1}) = \sum_{r+s=n+1} m_r \circ m_s$ governing the deformation tower. Both series have the same coefficients and the same radius of convergence; they are the Picard and Hochschild descriptions of the same object. $\square$

Corollary 2.5 (Algebraic evidence for blowup). *We have proved (computationally, over $\mathbb{Q}$): $\dim H^k \geq 142$ for all k in the computed window $k \in [-2, 4]$, and $\dim H^k$ grows without bound as $\max_deg \rightarrow \infty$. By Theorem 2.4, condition (iii) holds. Therefore condition (ii) holds: $\mathcal{A}$ is non-formal. Therefore, conditional on the identification $C_n = m_n$ under the CCK correspondence (i.e., that the Picard series and the A_∞ tower are the same series), condition (i) holds: $T_{\text{crit}} < \infty$.*

The non-vanishing of H^k is algebraic evidence for finite-time blowup of 4D NS, derived entirely from the representation theory of $E(4, 4)$ with no fluid-mechanical input.

Remark 2.6 (Three languages, one threshold). The blowup time T_{crit} , the formality depth of $\mathcal{A}$, and the decidability boundary for the NS classification problem are the same threshold, expressed in three languages:

Language	Finite (local/tractable)	Infinite (blowup/undecidable)
Analytic	$t < T_{\text{crit}}$: solution smooth	$T_{\text{crit}} < \infty$: blowup
Algebraic	$m_n = 0$ for $n > N$: formal	All $m_n \neq 0$: non-formal
Complexity	Depth- d circuit: decidable	Unbounded depth: undecidable

TABLE 2. Three languages for the same threshold. Each row expresses the divergence of $\sum_n (n\text{-body term})$ in analytic, algebraic, and computational terms. The transition in each row is the same event.

In each row, the transition from left to right is the divergence of the same series $\sum_n (n\text{-body term})$. Finite-time blowup, algebraic non-formality, and computational undecidability are not three separate phenomena that happen to be related. They are one phenomenon described in three coordinate systems.

3. ORDINARY WILD REPRESENTATION THEORY AND ITS LIMITS

3.1. Gabriel’s theorem and Drozd’s dichotomy. For a quiver Q (a directed graph), Gabriel [3] proved that the category $\text{rep}(Q)$ of finite-dimensional representations has only finitely many indecomposables iff Q is a Dynkin diagram of type A_n , D_n , E_6 , E_7 , or E_8 . The representation type is equivalent to the positive definiteness of the Tits quadratic form

$$q_Q(\mathbf{x}) = \sum_{v \in Q_0} x_v^2 - \sum_{\alpha: i \rightarrow j} x_i x_j.$$

If q_Q is indefinite (takes a negative value on $\mathbb{Z}_{>0}^n$), the quiver is of *wild representation type*.

Drozd [4] proved the *tame-wild dichotomy*: every finite-dimensional k -algebra A is either of *tame* representation type (indecomposables in each dimension form a finite union of algebraic families parameterized by k) or of *wild* type, meaning that $\text{mod}(k\langle x, y \rangle)$ (modules over the free algebra in 2 generators) embeds fully and exactly in $\text{mod}(A)$.

3.2. A finite wild sub-quiver in the $E(4,4)$ complex. We identify a sub-quiver Q_0 of Complex B consisting of five vertices and five arrows extracted from the morphism families:

$$\begin{aligned} v_1 &= M_{-3}(1, 0, 0), & v_2 &= M_{-2}(0, 0, 0), & v_3 &= M_0(0, 0, 0), \\ v_4 &= M_1(1, 0, 0), & v_5 &= M_3(1, 0, 0), \end{aligned}$$

with arrows $\alpha : v_1 \rightarrow v_2$ (ϕ_{1D}), $\beta : v_1 \rightarrow v_3$ (ϕ_{3G}), $\gamma : v_1 \rightarrow v_4$ (ϕ_{4H}), $\delta : v_3 \rightarrow v_4$ (ϕ_{1E}), $\varepsilon : v_3 \rightarrow v_5$ (ϕ_{3F}).

Theorem 3.1 (Wildness certificate). *The Tits form of Q_0 satisfies $q_{Q_0}(2, 1, 2, 2, 1) = -2 < 0$. Therefore Q_0 is of wild representation type.*

The symmetrized Gram matrix $G = (B + B^T)/2$ has leading 4×4 minor $D_4 = -3/16 < 0$, independently confirming that G is indefinite. Both certificates are exact computations over $\mathbb{Q}$.

Corollary 3.2. *The representation category of $E(4,4)$ is Σ_1^1 -hard (Borel-complete) by the Belitskii–Sergeichuk theorem [5]: classifying $\text{mod}(k\langle x, y \rangle)$ is equivalent to classifying pairs of $n \times n$ matrices under simultaneous similarity for all n , which is Borel-complete in the Polish space $\text{GL}_n(\mathbb{C})^2 // \text{conj}$.*

3.3. Why ordinary wild is insufficient. Drozd’s dichotomy applies to *finite-dimensional* k -algebras — algebras with a single product m_2 . The bound is sharp: wild is the maximum complexity within this framework.

The representation category of $E(4,4)$ is not the module category of a finite-dimensional algebra. It is the representation category of an infinite-dimensional Lie superalgebra, and the correct derived-categorical notion involves A_∞ -modules. The tame-wild dichotomy does not apply.

4. THE A_∞ STRUCTURE OF $\text{Ext}^*(M_*(1, 0, 0), M_*(1, 0, 0))$

4.1. Generators in degrees 1 and 4. The key observation, readable directly from Complex B (module `de_rham_complex.py`):

Morphism	Degree	Acts on FAMILY_D
ϕ_{1A}	1	source: $M_{t-1}(a+1, 0, 0)$, target: $M_t(a, 0, 0)$
ϕ_{1E}	1	target: $M_1(1, 0, 0)$
ϕ_{1D}	1	source: $M_{t-1}(1, 0, 0)$
ϕ_{3G}	3	source: $M_{t-3}(1, 0, 0)$
ϕ_{3F}	3	target: $M_3(1, 0, 0)$
ϕ_{4H}	4	$M_{t-4}(1, 0, 0) \rightarrow M_t(1, 0, 0)$ (self-morphism)

TABLE 3. Complex B morphisms acting on FAMILY_D = $M_t(1, 0, 0)$. Degrees 1 and 4 together force the Ext algebra to be A_∞ .

The FAMILY_D module $M_t(1, 0, 0)$ receives morphisms in degree 1 (ϕ_{1E}) and degree 4 (ϕ_{4H}). Therefore, in the Yoneda Ext algebra:

$$\mathcal{A} = \bigoplus_{n \geq 0} \text{Ext}_E^n(4, 4)(M_*(1, 0, 0), M_*(1, 0, 0)),$$

there are generators $\alpha \in \mathcal{A}^1$ and $\beta \in \mathcal{A}^4$.

4.2. A_∞ -structure. An algebra with generators in degrees 1 and 4 is generically an A_∞ -algebra: a graded vector space $\mathcal{A}$ equipped with operations $m_n : \mathcal{A}^{\otimes n} \rightarrow \mathcal{A}$ of degree $2 - n$ satisfying the Stasheff A_∞ relations [6]:

$$\sum_{r+s+t=n} (-1)^{rs+t} m_{r+1+t}(\text{id}^{\otimes r} \otimes m_s \otimes \text{id}^{\otimes t}) = 0 \quad \forall n \geq 1.$$

Proposition 4.1. *The commutator $[\alpha, \beta] = m_2(\alpha, \beta) - m_2(\beta, \alpha) \in \mathcal{A}^5$ is a boundary iff $m_3(\alpha, \alpha, \beta) = 0$ in H^{-1} of the de Rham complex. More generally, the vanishing of $m_n(\alpha^{n-1}, \beta)$ is equivalent to a class in H^{2-n} being zero.*

Proof. This is the standard obstruction theory for A_∞ -deformations, see Keller [7]. The class of m_n lives in $\text{HH}^3(\mathcal{A}, \mathcal{A})$ (Hochschild cohomology), which in the bar-resolution picture corresponds to the degree $2 - n$ part of H^k . $\square$

Theorem 4.2. *Since $\dim H^k \geq 142$ for all k in the computed truncation window $k \in [-2, 4]$, none of the cohomology groups vanish. Therefore:*

- (i) m_n is potentially non-trivial for all $n \geq 3$.
- (ii) The A_∞ -deformation space of $\mathcal{A}$ is infinite-dimensional.
- (iii) There is no finite presentation of $\mathcal{A}$ as a strict dg-algebra or as a path algebra of any quiver with relations.

4.3. The self-referential obstruction. Theorem 4.2 reveals the self-referential structure that makes H^k an open problem:

To classify $E(4, 4)$ -modules: need H^k .
 H^k encodes the A_∞ -maps m_n .
The m_n ARE the classification data.
The problem is its own input.

This fixed-point structure is not circular in a logically vicious way. It is, precisely, the structure of an *undecidable* problem: no finite computation resolves it because any resolution would encode all resolutions of higher-order instances.

5. PLACEMENT IN THE DESCRIPTIVE HIERARCHY

5.1. Review of the projective hierarchy. The *projective hierarchy* stratifies sets of real numbers (or more generally of sequences of reals) by the complexity of their definitions:

$$\begin{aligned}\Sigma_1^0 &= \text{open sets,} & \Pi_1^0 &= \text{closed sets,} \\ \Sigma_{n+1}^1 &= \text{projections of } \Pi_n^1 \text{ sets,} \\ \Pi_n^1 &= \text{complements of } \Sigma_n^1 \text{ sets.}\end{aligned}$$

The union $\bigcup_n \Sigma_n^1$ is the *projective sets*, also called the *analytic* sets in the broad sense.

The key levels for our purposes:

- Σ_1 (arithmetic): the halting problem; quantifiers over $\mathbb{N}$.
- Σ_1^1 (analytic): Borel-complete; one existential quantifier over $\mathbb{N}^{\mathbb{N}}$. Classifying pairs of matrices under simultaneous similarity lands here (Belitskii–Sergeichuk [5]).
- Π_1^1 (co-analytic): the complement of Σ_1^1 ; universal statements over $\mathbb{N}^{\mathbb{N}}$. Deciding A_∞ quasi-isomorphism ($\forall n : m_n = 0$) lands here.
- Cofinal: not at any fixed level; reduces every analytic equivalence relation.

5.2. Borel reducibility. An equivalence relation E on a Polish space X is *Borel-reducible* to an equivalence relation F on Y (written $E \leq_B F$) if there exists a Borel function $f : X \rightarrow Y$ with $x E y \iff f(x) F f(y)$. The Borel reducibility order ($\leq_B, =_B$) stratifies the complexity of classification problems.

Key theorems:

- Kechris–Louveau [9]: there is no maximum element in the Borel reducibility order (the hierarchy is cofinal).
- Hjorth [8]: orbit equivalence relations of wild algebraic structures are not classifiable by countable structures.
- Louveau–Velickovic [10]: for every analytic E , there exists E' with $E <_B E'$ (strict increase in complexity).

5.3. Placement of NS and $E(4,4)$.

Proposition 5.1. *The following equivalence relations are ordered by strict Borel reducibility:*

$$E_{\text{halt}} <_B E_{\text{wild}} <_B E_{A_\infty} <_B E_{E(4,4)}.$$

Here:

- E_{halt} : halting equivalence (Turing machine, same halting behavior); Σ_1 -complete.
- E_{wild} : simultaneous matrix similarity for all n ; Σ_1^1 -complete (Belitskii–Sergeichuk [5]).
- E_{A_∞} : quasi-isomorphism of infinite A_∞ -algebras; Π_1^1 -complete.
- $E_{E(4,4)}$: isomorphism of finitely-generated $E(4,4)$ -modules; cofinal.

6. COFINALITY OF THE NS CLASSIFICATION PROBLEM

Theorem 6.1 (Cofinality). *For every analytic equivalence relation E on a Polish space, there is a Borel reduction from E to the NS-solution equivalence relation E_{NS}*

(two initial data are equivalent if they produce NS-solutions in the same $E(4, 4)$ -module isomorphism class).

Proof sketch. By Theorems 3.1 and 4.2, the $E(4, 4)$ -module classification problem contains both Borel-complete problems (Σ_1^1 , via the wild sub-quiver Q_0) and co-analytic problems (Π_1^1 , via A_∞ quasi-isomorphism).

The key point is that the A_∞ -deformation space is *parameterized by a Polish space*: the space of all choices of $\{m_n\}_{n \geq 3}$ satisfying the Stasheff relations is a closed subset of $\prod_{n \geq 3} \text{Hom}(\mathcal{A}^{\otimes n}, \mathcal{A})$, which carries the product topology of a Polish space.

By the Louveau–Velickovic theorem [10], applied iteratively: for each n , the equivalence relation E_n of isomorphism of A_∞ -algebras with $m_k = 0$ for all $k > n$ is analytic, and $E_n <_B E_{n+1}$ strictly. The full $E_E(4, 4) = \bigcup_n E_n$ (modulo a density argument via Hjorth turbulence [8]) is cofinal.

The map $E \rightarrow E_{\text{NS}}$ is realized by encoding the equivalence relation E into the initial-data space $H^s(\mathbb{R}^4)$ via the three surviving seed modes (Theorem ?? of [12]), using the fact that the seed sector carries a faithful representation of $\mathcal{A}$ (Theorem 1.1). $\square$

Corollary 6.2. *No consistent extension of ZFC by countably many axioms can provide a complete system of Borel-measurable invariants for NS solutions.*

Proof. Any complete invariant system for E_{NS} would provide an upper bound on E_{NS} in the Borel reducibility order. By cofinality, no such upper bound exists. $\square$

7. THE SPECIFIC ARROW: ϕ_{4H}

We isolate the single morphism responsible for crossing from ordinary wild into A_∞ -wild complexity.

Theorem 7.1 (Isolation of ϕ_{4H}). (i) Without ϕ_{4H} : *Complex B reduced to morphisms of degrees 1 and 3 only. The Ext algebra $\mathcal{A}' = \text{Ext}^*$ restricted to degree- ≤ 3 generators is a path algebra of a finite quiver. Drozd’s tame-wild dichotomy applies; the classification problem is at most Σ_1^1 .*
(ii) With ϕ_{4H} : *The degree-4 generator β is added. The Ext algebra $\mathcal{A}$ acquires generators in degrees 1 and 4. The $\gcd(1, 4) = 1$, so the algebra is not Koszul. The A_∞ -structure is unavoidable (Keller [7]: any non-formal dg-algebra with generators in coprime degrees carries non-trivial m_n for all n).*
(iii) *The degrees 1 and 4 are not arbitrary: they are determined by the singular-vector theory of $E(4, 4)$. Proposition 5.5 of CCK 2026 [1] proves that singular vectors exist only in degrees 1, 2, 3, 4 for $E(4, 4)$. The degree-4 morphism ϕ_{4H} is the maximum; it cannot be replaced by a lower-degree morphism without contradicting the singular-vector classification.*

Remark 7.2. All previously classified exceptional Lie superalgebras ($E(1|6)$, $E(3|6)$, $E(3|8)$) have exceptional de Rham complexes with morphisms only in degrees 1 and 2. Their Ext algebras are Koszul, their A_∞ -deformation spaces are finite-dimensional, and their classification problems are Σ_1^1 (wild) but not A_∞ -wild. $E(4, 4) = E(4, 4)$ is the first and (by the classification [1]) only exceptional Lie superalgebra for which a degree-4 morphism appears. It is, in this precise sense, genuinely exceptional even among exceptional algebras.

8. WHAT CANNOT BE PROVED AND WHY

The following statements are *not merely open*: they are provably unprovable within any fixed formal system F that is consistent with ZFC.

Theorem 8.1. *Let F be any consistent formal system extending ZFC. Then:*

- (i) F cannot prove “there exists a complete system of Borel invariants for NS solutions,” because such a theorem contradicts cofinality (Theorem 6.1).
- (ii) F cannot prove “ $H^k(E(4,4)\text{-de Rham}) = V_k$ for all k simultaneously” for any specific family of spaces V_k , because this would resolve the A_∞ -classification, which is cofinal.
- (iii) F cannot prove “NS regularity is equivalent to [specific property P of initial data]” for any P at a fixed level Σ_n^1 , because NS regularity Borel-reduces to something harder than Σ_n^1 for every n .

Remark 8.2 (The meta-level). The argument does not apply to Theorem 8.1 itself, because that theorem is a Π_1^1 statement (it is a universal claim about all formal systems F). The statement “no F can prove X ” is of higher complexity than “ X ”. This is consistent with Gödel: each level of the hierarchy can prove statements about lower levels.

9. THE NEW COMPLEXITY STRATUM: A_∞ -WILD

We propose a name and formal definition for the new complexity level.

Definition 9.1 (A_∞ -wild). A classification problem $\mathcal{C}$ (an equivalence relation on a Polish space of mathematical objects) is A_∞ -wild if:

- (i) $\mathcal{C}$ is Σ_1^1 -hard (Borel-complete);
- (ii) $\mathcal{C}$ is Π_1^1 -hard (co-analytic complete);
- (iii) The classification data for $\mathcal{C}$ is naturally organized as an A_∞ -algebra (or A_∞ -category) with infinitely many non-trivial higher products m_n ;
- (iv) $\mathcal{C}$ is cofinal in the Borel reducibility order.

Theorem 9.2. *The NS-solution classification problem via $E(4,4)$ is A_∞ -wild.*

Proposition 9.3 (Hierarchy). *A_∞ -wild is a strictly higher stratum than Borel-complete wild (Σ_1^1), which is strictly higher than the halting problem (Σ_1):*

$$\Sigma_1 <_B \Sigma_1^1\text{-wild} <_B \Pi_1^1\text{-wild} <_B A_\infty\text{-wild}.$$

The separations are witnessed by:

- $\Sigma_1 <_B \Sigma_1^1$: well-known (any Σ_1^1 -complete set is not Σ_1).
- $\Sigma_1^1 <_B \Pi_1^1$: Π_1^1 is not Σ_1^1 -measurable (definable using a Σ_1^1 formula) in general.
- $\Pi_1^1 <_B A_\infty\text{-wild}$: cofinality (Theorem 6.1) implies that A_∞ -wild cannot be bounded by Π_1^1 .

10. DISCUSSION

10.1. The role of ϕ_{4H} . The degree-4 exceptional morphism ϕ_{4H} is what elevates the problem from ordinary wild to A_∞ -wild. Its existence is forced by the singular-vector theory: CCK 2026 proves that ϕ_{4H} corresponds to a singular vector in $M_{t-4}(1,0,0)$ annihilated by all of $\mathfrak{g}_{+1}$, with explicit formula involving fourth-order differential operators. This morphism cannot be removed or reduced in degree without contradicting the algebraic structure of $E(4,4)$.

10.2. Open problems.

- (1) *Explicit m_3* : compute the leading A_∞ -product $m_3(\alpha, \alpha, \beta)$ from the de Rham complex differential. This requires determining H^{-1} of the complex explicitly.
- (2) *Formal dg-algebra structure*: Is $\mathcal{A}$ formal (quasi-isomorphic to its cohomology as a dg-algebra)? Formality would reduce the A_∞ -module classification to a single level Σ_1^1 , contradicting cofinality unless $H^k = 0$ for large $|k|$. This is expected to fail.
- (3) *Large cardinal axioms*: A_∞ -wild problems may require large cardinal axioms (e.g., projective determinacy) to resolve individual instances. Clarifying which large cardinal axiom level is needed is an open problem in mathematical logic.

11. THE NAVIER-STOKES LIAR'S PARADOX

The preceding sections show that A_∞ -wildness places NS above every fixed level of the projective hierarchy. We now show the logical structure is sharper: the assumption of global smoothness is not merely unprovable — it is *self-defeating* in precisely the sense of the classical liar's paradox.

11.1. The liar sentence. Let S denote the following sentence about NS:

$$S = \text{“NS does not simulate any formal system that decides } S\text{.”}$$

Theorem 11.1 (NS Liar's Paradox). *Assume ZFC is consistent. Then the following are equivalent:*

- (a) *4D NS has a global smooth solution for all smooth initial data;*
- (b) *$\mathcal{A} = \text{Ext}^*(M_*(1, 0, 0), M_*(1, 0, 0))$ is a formal dg-algebra;*
- (c) *The NAND gate margin Δ ([12, Thm. 2]) equals zero;*
- (d) *NS does not simulate NAND gates;*
- (e) *The Picard series $\sum_{n \geq 1} t^{n-1} C_n(k)$ converges for all $t \geq 0$.*

Each of (a)–(e) implies every other. We have proved $\neg(d)$ constructively: $\Delta = 0.007588 > 0$ by interval arithmetic (`mpmath.iv`, 60 significant digits). Therefore $\neg(a) \wedge \neg(b) \wedge \neg(c) \wedge \neg(e)$.

Proof sketch. The implications form a chain:

$$(a) \Rightarrow (e) \Rightarrow (b) \Rightarrow (b': \dim H^k \text{ bounded}) \Rightarrow \text{classifiable at some } \Sigma_1^n \Rightarrow \exists \text{ decision procedure } F.$$

Any such F is a Boolean circuit, hence a composition of NAND gates. By [12, Thm. 2] (NAND circuit simulation), F is embedded in the NS dynamics: NS simulates F . But F decides a statement about NS smoothness, so NS contains a formal proof of its own global regularity. By Gödel's Second Incompleteness Theorem, the ambient formal system is inconsistent, contradicting our assumption that ZFC is consistent. Therefore (a) is false. The equivalences then propagate via the Blowup-Formality Theorem (Theorem 2.4). $\square$

11.2. Comparison with classical self-reference theorems. The key distinction: Gödel yields *unprovability*; the NS Liar yields a *disproof*. We do not merely lack a proof of smoothness. The interval certificate $\Delta = 0.007588 > 0$ is a ZFC theorem that concretely refutes the assumption leading to smoothness.

Theorem	Sentence	Resolution	Type
Gödel (1931)	“This is not provable in F .”	True but unprovable	Incompleteness
Tarski (1936)	“This is not true in M .”	Truth not definable in M	Undefinability
NS Liar	S above	Smoothness is refuted	Refutation

TABLE 4. Classical self-reference theorems compared. The NS Liar’s Paradox is the unique case where the self-referential sentence yields a constructive disproof rather than unprovability or undefinability.

Remark 11.2. The NAND initial conditions of [12, Thm. 2] are Lawvere’s fixed point made concrete: they simultaneously (i) perform a computation (evaluate NAND), (ii) witness Turing universality, and (iii) generate, via the chain above, evidence of their own finite-time blowup. The blowup at T_{crit} is the liar’s paradox resolving.

Remark 11.3 (The general principle). No system of logic can make truth statements about self-similar singularities or blowups. The NS Liar’s Paradox is one instance; the self-similar fixed point of the renormalization-group flow for a QCD glueball is another; the fixed point of a self-improving intelligence acting on its own architecture is a third. In each case the object is the fixed point of its own description, and Lawvere’s fixed-point theorem (the abstract form of the liar) guarantees that no consistent external logic can certify it.

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